IoT4Ag
The Internet of Things for Precision Agriculture
a U.S. NSF Engineering Research Center

Introduction to Grafana Visualization with Data Sources and Dashboards

Faik Kerem Ors, Yaguang Zhang

Penn University of Pennsylvania | Purdue University | University of California Merced | UF University of Florida | ASU Arizona State University | NSF NSF ERC

What is Grafana?

- Open-source Visualization Platform
  - Time-series analysis, long-running system behavior
- Data sources
- Dashboards
  - Panels/Graphs
- Examples:
  - AgLab - ag.publicvm.com
  - SPRING Portal
  - My Grafana Cloud



Source: https://grafana.com/oss/grafana/

Python vs Grafana

Python

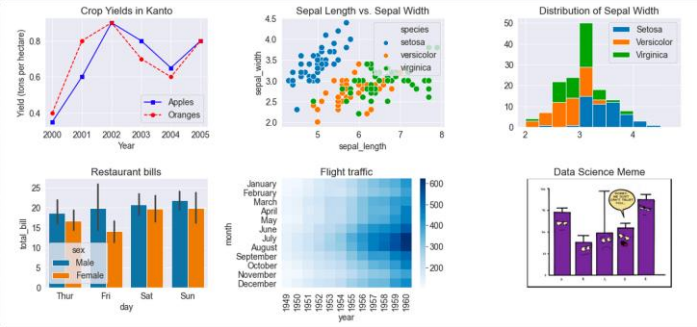
Maximum flexibility (code-first visuals + analysis)

Best for

- Exploratory analysis, stats, ML evaluation
- Custom or publication-quality charts
- Complex data shaping (pandas/polars)

Watch-outs

- Always-on dashboards require extra engineering/hosting
- Alerting & access control are mostly DIY



Source: https://python.plainenglish.io/data-visualization-with-python-8ce66744ccbd

Grafana

Fast dashboards (monitoring + sharing + alerting)

Best for

- Real-time / time-series monitoring dashboards
- Sharing/Embedding (URLs, roles, templated filters)
- Alerts on metrics/logs with notifications

Watch-outs

- Less flexible for niche chart types / heavy annotation
- Complex transforms often belong upstream (SQL/ETL)



Source: https://grafana.com/oss/grafana/

Rule of thumb: 'monitor now' → Grafana | 'analyze why' → Python | often use both

Quiz/Activity

Penn University of Pennsylvania | Purdue University | University of California Merced | UF University of Florida | ASU Arizona State University | NSF NSF ERC

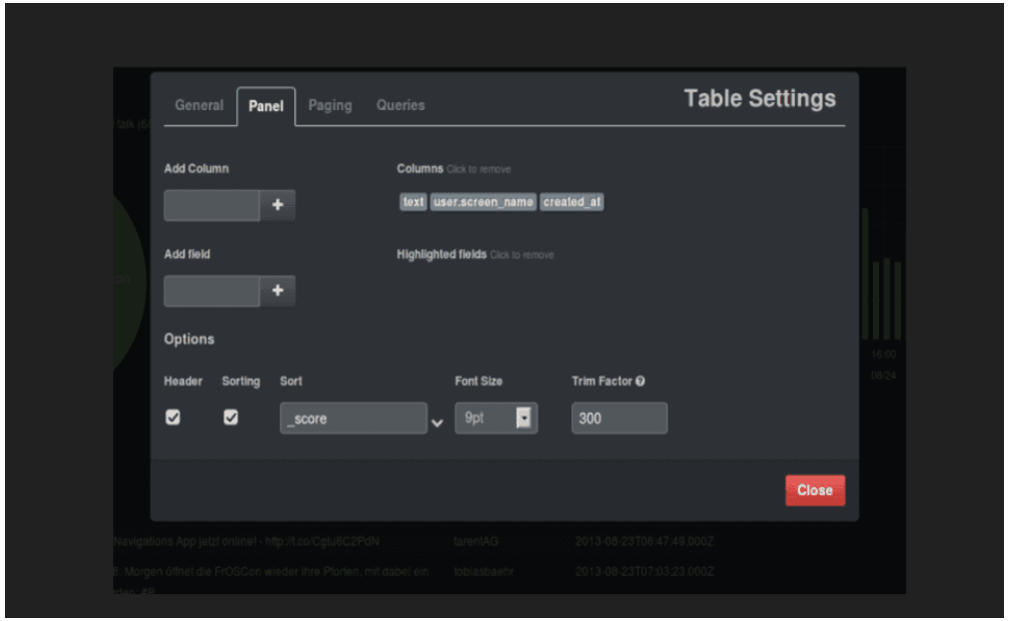
Grafana History

Source: https://grafana.com/blog/the-mostly-complete-history-of-grafana-ux/

Origin story (pre-2014)

- Inspired by Kibana 3 drag-and-drop dashboards
- Elastic Search + Kibana





Grafana History

Origin story (pre-2014)

- Inspired by Kibana 3 drag-and-drop dashboards
- Elastic Search + Kibana



Key UX milestones

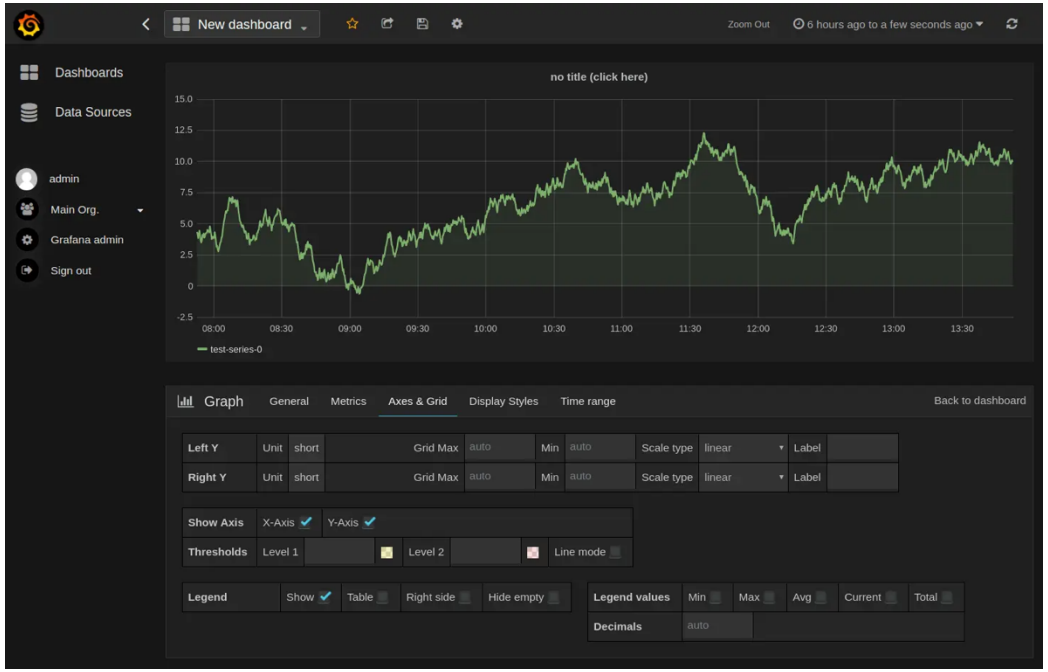
2014 — v1: full-screen panel edit, immediate feedback, readable query builder

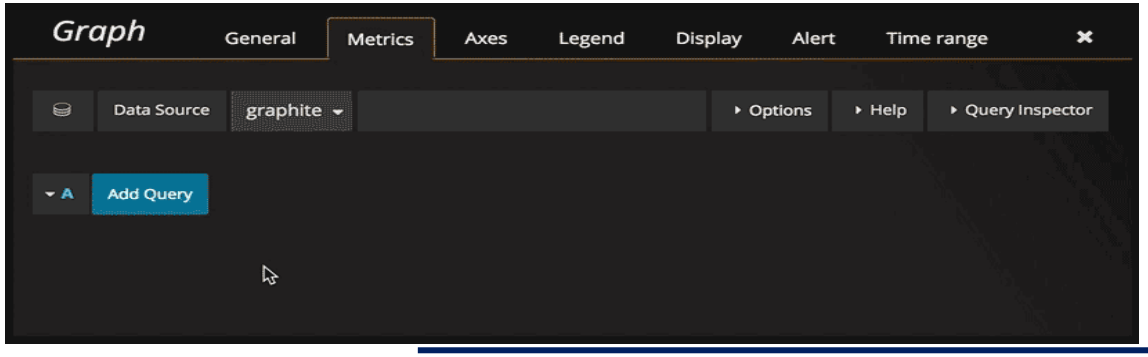


Grafana History

Key UX milestones

2014 — v1: full-screen panel edit, immediate feedback, readable Graphite query builder
2015 — v2: Go backend, login and save dashboards; datasources via UI
2016 — v3/4: plugin system, adds alerting
2018 — v5: improved side menu, polished light theme
2019 — v6: switch graph types without recreating panels, improved edit UX



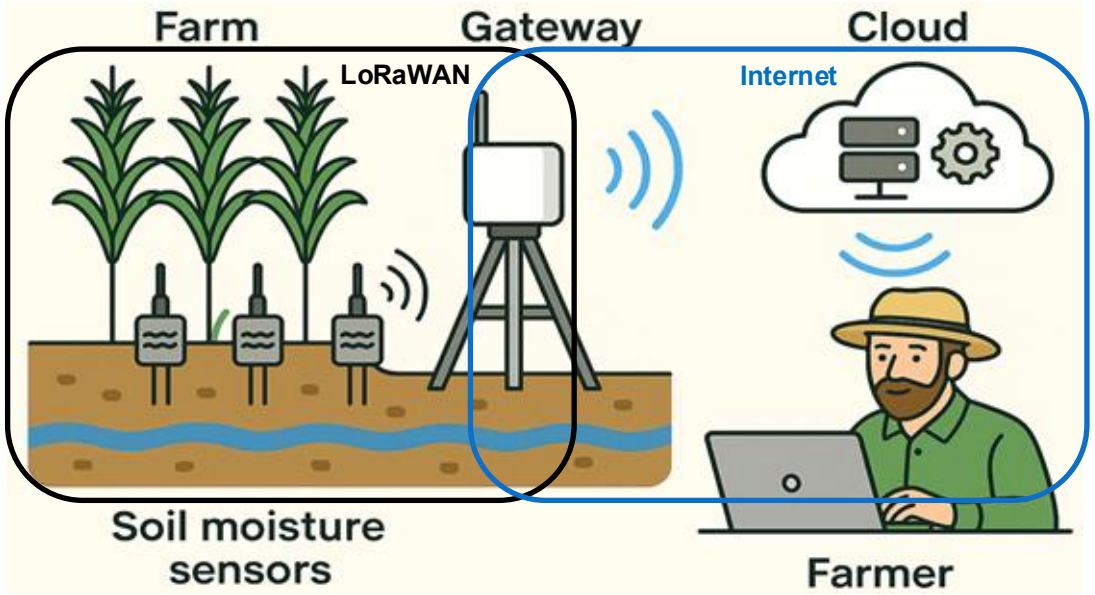




Quiz/Activity

Penn University of Pennsylvania | Purdue University | University of California Merced | UF University of Florida | ASU Arizona State University | NSF NSF ERC

Data Flow into Grafana





Dragino SE01-LB

Soil Moisture and Temperature



Atmos 41 Gen 2 Weather Station

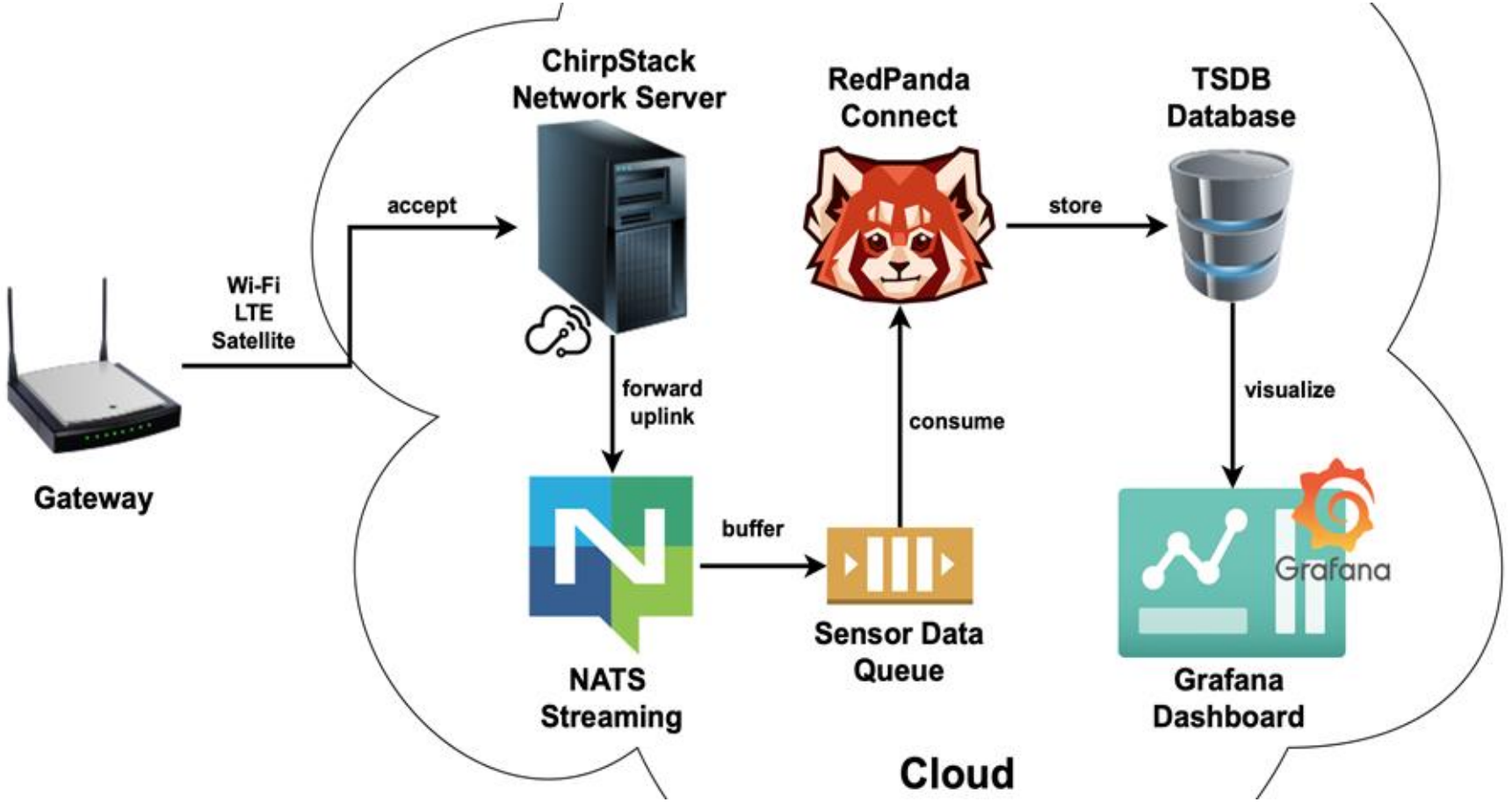


Tektellic Clover



RAKwireless WisGate Edge Pro 8 Channel

Data Flow into Grafana



Quiz/Activity

Steps to Build Grafana Dashboards (Demo)

1. Store your data somewhere: File or database
2. Connect your data source to Grafana
3. Create a new dashboard
4. Create a panel inside the dashboard
5. Build a query to grab data from your data source
6. Select the appropriate graph for your data







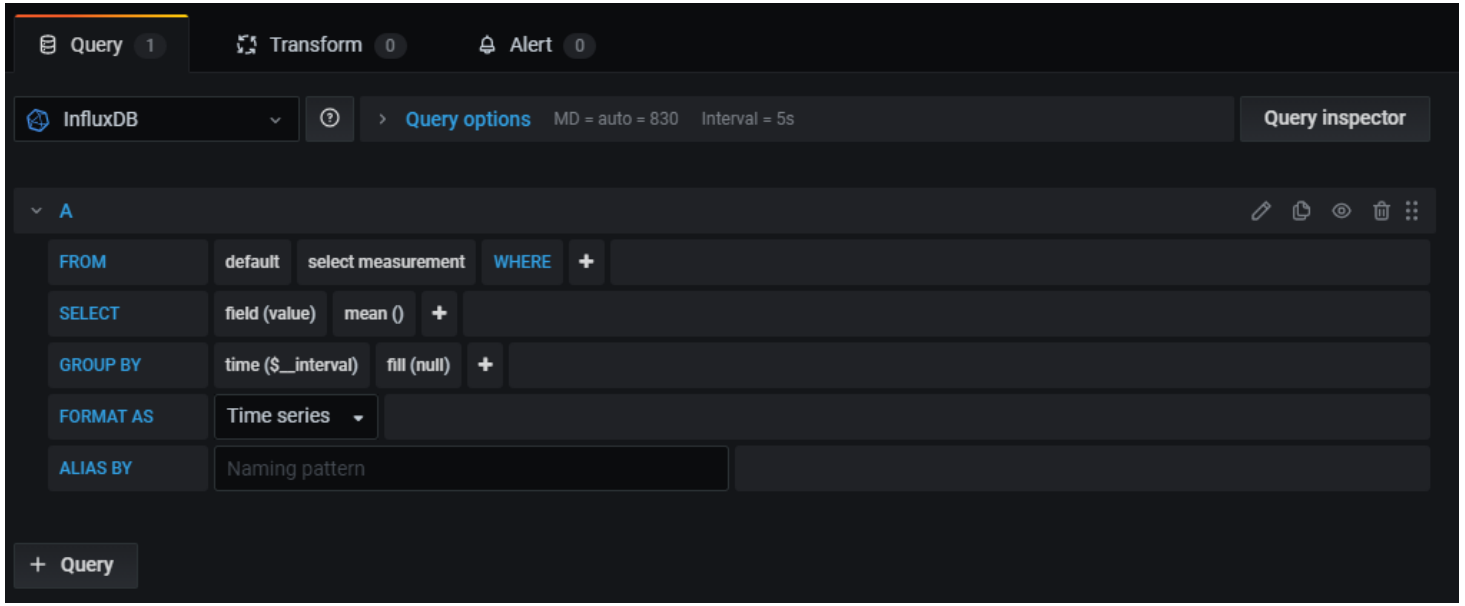
Source: https://grafana.com/oss/grafana/

Grafana Data Sources

- Grafana comes with built-in support for many *data sources*.
- Each data source comes with a *query editor*, which formulates custom queries according to the source's structure.

- Because of the differences between query languages, each data source query editor looks and functions differently.

- Special data sources
  - Grafana, Mixed, Dashboard



Source: https://grafana.com/docs/grafana/latest/datasources/

Grafana Data Sources

- These built-in core data sources are also included in the Grafana documentation

- Alertmanager
- AWS CloudWatch
- Azure Monitor
- Elasticsearch
- Google Cloud Monitoring
- Graphite
- InfluxDB
- Jaeger
- Loki
- Microsoft SQL Server (MSSQL)
- MySQL
- OpenTSDB
- PostgreSQL
- Prometheus
- Pyroscope
- Tempo
- Testdata
- Zipkin



Source: https://grafana.com/docs/grafana/latest/datasources/

Grafana Data Sources: CSV Files

- CSV Plugin for Grafana
  - Outdated, maintenance mode, no new features will be added.
- Infinity Plugin
  - Supports multiple file types such as CSV, JSON and XML.
  - No local csv file option unless served in a self-managed HTTP server.



Source: https://grafana.com/media/products/assets/grafana/Grafana_Plugins.png



Source: https://grafana.com/grafana/plugins/yesoreyeram-infinity-datasource/

Grafana Data Sources: Files (Demonstration)

- Example 1: Use CSV Plugin for Grafana to build a dashboard
- Example 2: Use Infinity Plugin to build a dashboard

Quiz/Activity

Databases

- Files break down because...
  - Slow to query at scale
  - No efficient time-range filtering
  - Concurrency and integrity issues (Multi-user modifications, rollback issues)
  - No indexing (Full scan/search)

- Database management systems
  - Fast queries with indexes
  - Bult-in aggregation and filtering (SELECT, WHERE, GROUP BY)
  - Consistent schema
  - Avoids data corruption issues due to modifications and rollbacks

Grafana Data Sources: Databases

- Built-in data sources
  - Different interfaces and query options
  - Elastic Search: A search and analytics engine, built for fast full-text search and aggregations over large event/log datasets.
  - PostgreSQL: tables/rows with defined schema
- We will focus on PostgreSQL (TimescaleDB)
- Let's see what a real database look like!
  - Database, Tables, Rows, Columns, Values

Source: https://grafana.com/oss/grafana/

Databases and Query Building with SQL

- SQL: Structured Query Language
  - Standard language for storing, manipulating and retrieving data in databases.

- **SELECT** * **FROM** <table> **WHERE** <condition>;

- Examples:

```
SELECT crop_name
FROM crops
WHERE crop_type = 'vegetable';
```

```
SELECT COUNT(*) AS crop_count
FROM crops;
```

- You don't need be an expert, feel free to use AI ! ☺

Databases and Query Building with SQL

- SQL: Structured Query Language
  - Standard language for storing, manipulating and retrieving data in databases.

- **CREATE TABLE** <table> (**FIELDS**);

- Examples:

```
CREATE TABLE students (
    student_id INTEGER PRIMARY KEY,
    student_name VARCHAR(100) NOT NULL
);
```

- You don't need be an expert, feel free to use AI ! ☺

Grafana Data Sources: Databases (Demo)

- Example: Use PostgreSQL data source to build a dashboard
  - Generate SQL file from CSV file
  - Connect data source
  - Create a new dashboard and/or panel
  - Build SQL queries with different graph types
    - Grafana SQL macros
      - $__timeFilter(column)
      - $__timeFrom() / $__timeTo()

Source: https://grafana.com/oss/grafana/

Grafana Data Sources: Pros and Cons

Files

Simple for storing (small) data quickly

Pros

- Simple to create and share
- Human-readable and works with many tools
- Good for small, static datasets

Cons

- Slow to query at scale (no indexing, full file scans)
- No efficient time-range filtering
- Concurrency and integrity issues in multi-user environments

Databases

Answering questions from a large amount of data quickly

Pros

- Fast queries with indexes
- Built-in filtering and aggregation (SELECT, WHERE, GROUP BY)
- Consistent schema and validation
- Safer multi-writer ingestion, avoiding corruption
- Supports rollbacks, history and auditing through transactions

Cons

- More effort for setting up (hosting, backups, monitoring)
- Careful schema design and migrations

Source: https://grafana.com/oss/grafana/

Rule of thumb: ‘store small data quickly’ → Files | ‘answer questions from large amount of data’ → Databases

Quiz/Activity

Thank You!